

1. Administrative & Study Identification

Full Study Title: A Single-arm, Open-label, Phase II Study of Sabatolimab in Combination With Azacitidine and Venetoclax in Adult Participants With High or Very High Risk Myelodysplastic Syndromes (MDS) as per IPSS-R Criteria **Short Title/Acronym:** STIMULUS-MDS3 **Protocol Number:** CMBG453B12203 **NCT Number:** NCT04812548 **Posting ID:** 743068943421786977 **Sponsor/Company:** Novartis **Investigational Product:** Sabatolimab (MBG453), Azacitidine (Trade names: Vidaza, Onureg; ATC Code: L01BC07), and Venetoclax **Phase of Development:** Phase II **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety of sabatolimab administered in combination with azacitidine and venetoclax (Part 1: rule out excessive dose-limiting toxicity) and to determine the complete remission (CR) rate of the triplet regimen (Part 2). **Secondary Objectives:** To evaluate efficacy measures including morphological complete remission (mCR), overall response rate (ORR) with hematologic improvement (HI), and RBC/Platelet transfusion independence.

2.2 Background & Rationale Patients with High-Risk Myelodysplastic Syndromes (HR-MDS) who are unfit for intensive chemotherapy or hematopoietic stem cell transplant (HSCT) have poor outcomes with hypomethylating agents (HMAs) alone. Sabatolimab is a novel immuno-myeloid therapy that binds to TIM-3 on immune cells and leukemic cells, facilitating antileukemic immune activation and potentially impeding self-renewal of leukemic stem cells. This trial explores whether the triplet therapy of sabatolimab combined with an HMA (azacitidine) and a BCL-2 inhibitor (venetoclax) can provide safe, durable clinical benefits and survival advantages in this high-risk patient population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Trial Status: TERMINATED
Targeted Enrollment: 20 participants. *(Note: Originally planned for ~76 participants—~18 in Part 1 safety run-in and ~58 in Part 2 expansion. Due to the decision by Novartis to halt recruitment*

at the end of the Safety run-in, the study enrolled participants in Part 1 only and did not meet the target of 76.) **Number of Sites:** International, Multicenter **Estimated Study Start:** 31-May-2021 **Estimated Primary Completion:** 08-May-2023

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Morphologically confirmed diagnosis of myelodysplastic syndrome (MDS) based on the 2016 WHO classification. 2. Prognostic Risk Categories, based on the revised International Prognostic Scoring System (IPSS-R): Very high (> 6 points) or High (> 4.5-6 points). 3. Unsuitable/Not immediately eligible for treatment with intensive chemotherapy or a hematopoietic stem-cell transplant (HSCT) at the time of screening.

4.2 Exclusion Criteria 1. Previous first-line treatment for very high risk or high risk MDS with any antineoplastic agents, approved or investigational (including chemotherapy, lenalidomide, or HMAs). 2. *Exception:* A single cycle of HMA treatment started prior to enrollment is allowed. 3. Prior exposure to TIM-3 directed therapy or any BCL-2 inhibitor (including venetoclax) at any time.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Part 1 consisted of 2 sequential cohorts: Sabatolimab 400 mg (Cohort 1) or 800 mg (Cohort 2) administered every 4 weeks (Q4W), combined with a fixed dose of venetoclax and azacitidine. **Route of Administration:** Intravenous (Sabatolimab, Azacitidine), Oral (Venetoclax). **Duration of Treatment:** 28-day treatment cycles administered until disease progression, unacceptable toxicity, or trial termination.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for cytopenias and infections. **Prohibited:** Concurrent administration of other systemic antineoplastic agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Bone Marrow Biopsy, IPSS-R calculation
Baseline	Day 0	Registration, First Dose of Triplet Therapy, Safety Labs
Interim	Day 1 of each 28-day Cycle	Safety Labs, Efficacy Assessment (Bone Marrow Aspirate/Biopsy), AE monitoring
End of Study	Follow-up	Final Vital Signs, Exit Assessment, Survival follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- 1. Rate of Dose Limiting Toxicities (DLTs) - All Grades (Safety run-in Patients Only):** Evaluation of dose-limiting toxicities during the DLT observation period.
- 2. Percentage of Participants (Receiving 800mg Sabatolimab) Achieving Complete Remission (CR) Per Investigator Assessment:** Defined as bone marrow blasts $\leq 5\%$, hemoglobin ≥ 10 g/dL, platelets $\geq 100 \times 10^9/L$, neutrophils $\geq 1.0 \times 10^9/L$, absence of blasts in peripheral blood.

7.2 Secondary Outcomes

- 1. Percentage of Subjects Achieving a Complete Remission (CR) + Morphologic Complete Remission (mCR):** Safety run-in (Part 1).
- 2. Overall Response Rate (ORR) of Participants Who Achieved Hematologic Improvement (HI) or Better as Best Response.**
- 3. Percentage of Participants Who Are RBC/Platelets Transfusion Independent.**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of the incidence and severity of AEs and Serious AEs (SAEs). **Serious Adverse Events (SAEs):** Standard reporting timeline to the sponsor and regulatory authorities. **Data Monitoring Committee (DMC):** Safety data from Cohort 1 were reviewed by a safety committee to ensure the regimen was safe prior to opening Cohort 2.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients receiving at least one dose of study drug) and Efficacy Set (Intent-to-Treat patients). **Sample Size Justification:** The safety run-in (Part 1) planned for ~6 patients in the 400 mg cohort and ~12 patients in the 800 mg cohort to adequately assess DLTs. Expansion (Part 2) originally targeted ~58 patients to provide sufficient power to evaluate CR, but the study was terminated early. **Handling of Missing Data:** Survival data missing due to loss to follow-up were censored at the date of last contact.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote sponsor monitoring visits to ensure protocol adherence and safety recording. **Audit Trail:** Data captured via electronic Case Report Forms (eCRF) with standardized data clarification processes.

11. Study Identifiers & Governance

Primary ID: CMBG453B12203 **Posting ID:** 743068943421786977 **Registry Number:** NCT04812548 **Sponsor/Lead Organization:** Novartis **Study Title:** STIMULUS-MDS3 **Official Regulatory Title:** A Single-arm, Open-label, Phase II Study of Sabatolimab in Combination With Azacitidine and Venetoclax in Adult Participants With High or Very High Risk Myelodysplastic Syndromes (MDS) as per IPSS-R Criteria

12. Clinical Framework (MDS Specific)

Primary Condition: High or Very High Risk Myelodysplastic Syndromes (HR-MDS) **Condition Codes:** MeSH: D009190 | HPO: HP: 0002863 | SNOMED: 109995007 **Diagnosis Threshold:** IPSS-R Score > 4.5 points **Medical Methodology:** Triplet Therapy (TIM-3 inhibitor + HMA + BCL-2 inhibitor) **Optimization Target:** Complete Remission (CR) and prevention of leukemic progression

13. Study Procedures & Methodology

Management Pathway: Intensive triplet intervention pathway for higher-risk MDS **Education Protocol (HCP):** Site initiation visits and clinical protocol training on DLT management **Education Protocol (Patient):** Informed consent processes and dosing diaries for oral medications (venetoclax) **Quality Audit Mechanism:** Source data verification and dose-limiting toxicity monitoring

14. Observation & Follow-Up Schedule

Total Duration: Continued until disease progression or unacceptable toxicity **Onsite Visit Cadence:** Day 1 of each 28-day cycle, with more frequent visits during Cycle 1 and 2 **Remote Monitoring Frequency:** Per sponsor monitoring guidelines **Checklist Review Interval:** End of each treatment cycle

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				